

The Hong Kong University of Science and Technology

MECH 3620 – Aircraft Design

Spring 2025–2026

Course Syllabus

1 General Information

- **Instructor:** Zhenyu Gao (zhenyugao@ust.hk), Assistant Professor of MAE (Office: 2563)
- **Teaching Assistants:**
 - Muhammad Alfiyandy Hariansyah (mahariansyah@connect.ust.hk)
 - Fangyi Wu (fwuat@connect.ust.hk)
- **Lecture Time:** Wednesday & Friday 03:00 PM - 04:20 PM, Location: Room 4504, Lift 25-26
- **Tutorial Time:** Wednesday 11:00 AM - 11:50 AM, Location: LSK Room 1032
- **Instructor Office Hour:** Monday 15:30 PM - 16:30 PM, Location: Room 2563, Lift 27-28

2 Prerequisites

- **Prerequisites:** MECH 1907 – Introduction to Aerospace Engineering, MECH 3640 – Aerodynamics, MECH 3650 – Aircraft Structure
- **Corequisites:** MECH 3660 – Gas Turbines and Jet Propulsion, MECH 3670 – Aircraft Performance and Stability

3 Course Objectives

Welcome to MECH3620 Aircraft Design!!! This is a major requirement course in the BEng in Aerospace Engineering curriculum at HKUST (<https://mae.hkust.edu.hk/en/academics/undergraduate-program/beng-in-aerospace-engineering>). The course is a **project-based course**, where all students will work on a group project that involves computational modeling and programming. The course objectives are:

- Understand the **overall aircraft design process**, following the typical industry practice
- Understand the basic concepts of **design thinking and systems engineering** in a complex and multidisciplinary system such as aircraft
- Understand the **interrelations** between the different disciplines (e.g., aerodynamics, structures, stability, propulsion, etc) in an aircraft system
- Go through a simple conceptual design stage of an aircraft through a **course project**
- Learn how to use **programming and computing** tools to help solve real-world engineering problems
- Practice **scientific communication** skills (in English), both in written reports and in oral presentation
- Exercise good **teamwork** to achieve the course project's design objectives

4 Textbook and Reference Materials

Class notes ([slides](#) in PDF format and [Pluto notebooks](#)) will be provided and uploaded to Canvas per topic. In principle, the class notes should contain all the materials needed for the course. You will need to look for additional information when completing the individual term paper and the group project. Some [reference textbooks](#) are:

1. Daniel P. Raymer. Aircraft Design: A Conceptual Approach. AIAA, 5th edition, 2012.
2. Jack D. Mattingly, William H. Heiser, Keith M. Boyer, Brenda A. Haven and David T. Pratt. Aircraft Engine Design. AIAA, 3rd edition, 2018.

5 Course Assessments

Table 1: Course assessments and weights

Assessment	Weighting
Group Design Project	80%
Individual Take-home Quizzes	15%
Class Participation	5%
Total	100%

5.1 Group Design Project (80%)

- **Midterm project presentation:** 15%
- **Final project presentation:** 20%
- **Final project report:** 35%
- **Individual project report:** 10%

Working in a group of **around 4 students**, students will work on a conceptual aircraft design project. [Peer assessment](#) will be included in the final grading. The group project description and detailed instruction will be given in class, and the document will be posted on Canvas in due time. Students will use the in-house **Multidisciplinary Aircraft Design Education (MADE)** framework, called **AeroFuse**. The platform is developed using **Julia** programming language with Pluto notebook as interface. A thorough tutorial will be provided in the tutorial session. To be better prepared for the course (especially the term project), students are referred to the following important references:

- Julia programming language: <https://julialang.org/>
- Pluto notebook: <https://plutojl.org/>
- AeroFuse documentation: <https://hkust-octad-lab.github.io/AeroFuse.jl/stable/>
- AeroFuse platform repository: <https://github.com/HKUST-OCTAD-LAB/AeroFuse.jl>

Students are highly encouraged to **check them out early** (in the first few weeks of the class). Each student will also need to submit a [peer assessment](#), a [statement of contribution](#), and a [self-reflection](#) on learning experience. A template will be provided for this individual submission. Failing to submit peer assessments will incur a 10% penalty of the student's project grade.

5.2 Individual Take-home Quizzes (15%)

- **Quiz 1 – Aircraft design and sizing:** 5%
- **Quiz 2 – Aircraft configurations, aerodynamics, wing design:** 5%
- **Quiz 3 – Aircraft weight and balance, stability, structures, propulsion:** 5%

The quizzes will mainly test the aircraft design theory and knowledge discussed in the lectures. You will have 4 days to complete each take-home quiz. The detailed form of each quiz will be announced soon.

5.3 Class Participation (5%)

This includes enthusiasm, questions asked, questions answered, class attendance, punctuality, and in-class behavior.

6 Communication and Participation

- **Students' Talk Time at the beginning of each lecture:** We will open the first five minutes of each lecture to give opportunities for students to speak up. Please feel free to share your thoughts and give us feedback. You can basically share anything related to our class (and your opinions are important to us!!!), for example: (1) How we (the instructor and TAs) did in lecture and office hours (your complaints...); (2) What additional topics you'd like to learn in this course; (3) Questions or clarifications from the last lecture; (4) Recent news and discoveries in aviation and aerospace engineering; (5) Anything else you want to share!
- **Emails and Online Forum:** We (the instructor and TAs) are happy to answer ANY questions you have. When you have a question regarding the course materials or logistics, **please consider posting it to our online forum first** (Canvas). From our past experience, most questions are similar. When you post it to the online forum, we will do our best to provide an answer in a timely manner. The use of online forum will avoid repetitive Q&As (which makes our course more efficient) and benefit your classmates (very important!). For other questions (something more private or sensitive), you are welcome to email the instructor and TAs, or ask us during the office hours.

7 Collaboration Policy

- All take-home quizzes are **individual assignments**. You must work independently and are NOT allowed to discuss with your classmates. **Put academic integrity above anything else.** We are very strict about the honor code and practice **no tolerance** for any cheating. For more details, see HKUST Academic Honor Code (<https://registry.hkust.edu.hk/resource-library/academic-honor-code-and-academic-integrity>) and Declaration of Academic Integrity (<https://registry.hkust.edu.hk/resource-library/declaration-academic-integrity>).
- For **group projects**, certain communications between groups will be allowed, such as the general approaches, possible data sources, software, etc. However, each team should develop **some uniqueness** in your project. Please do differentiate your work from the others.
- For the use of **Gen AI tools** (e.g., large language models), you are NOT allowed to use AI tools to write the term paper and the project report for you (we will consider the use of necessary tools in grading). However, you can use these tools as “helpers” in smart ways, such as background search, brainstorming, and polishing sentences.

8 Tentative Course Schedule

Lec. No.	Date	Topic	Note
1	February 4	Course introduction	
2	February 6	Aircraft design overview	
3	February 11	Aircraft sizing: constraint analysis	Group project info out
4	February 13	Aircraft sizing: constraint analysis	Project group finalization
–	February 18	The second day of Lunar New Year	
5	February 20	Aircraft sizing: mission analysis & weight estimation	
6	February 25	Aircraft sizing: mission analysis & weight estimation	Quiz 1 out
7	February 27	Aircraft configurations	Quiz 1 due
8	March 4	Aircraft configurations	
9	March 6	Aerodynamic and wing design	
10	March 11	Aerodynamic and wing design	Quiz 2 out
11	March 13	Cockpit and fuselage layout design	Quiz 2 due
12	March 18	Midterm project review	Midterm presentation slides due
13	March 20	Midterm project review	
14	March 25	Midterm project review	
15	March 27	Weight and balance & stability analysis	
16	April 1	Weight and balance & stability analysis	
–	April 3	Good Friday	
17	April 8	Aircraft structures	
18	April 10	Propulsion system	Quiz 3 out
19	April 15	Empennage design & landing gear design	Quiz 3 due
20	April 17	Cost analysis	
21	April 22	Optimization and course review	
22	April 24	Guest lecture on novel aircraft design	
23	April 29	Final project presentation	Final presentation slides due
–	May 1	Labor Day	
24	May 6	Final project presentation	
25	May 8	Final project presentation	
–	May 15		Final reports due

9 Tentative Tutorial Schedule

Tut. No.	Date	Topic
1	February 4	Course project briefing + Introduction to Julia programming and GitHub
2	February 11	Introduction to AeroFuse platform + programming tutorials
–	February 18	The second day of Lunar New Year
3	February 25	Drag polar and constraint analysis (matching chart)
4	March 4	Preliminary weight estimation
5	March 11	Preliminary wing design
6	March 18	Midterm project consultation
7	March 25	Midterm project consultation
8	April 1	Preliminary weight and balance
9	April 8	Structural analysis + propulsion system
10	April 15	Final project consultation
11	April 22	Final project consultation
12	April 29	Final project consultation